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RESEARCH ARTICLE

Impact of Data Leakage in Vibration Signals Used for Bearing Fault Diagnosis

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ABSTRACT Bearing fault diagnosis is a well-developed field and an active area of research in which the combination of model-free machine learning techniques with vibration data has become a popular approach. However, vibration data from rotating machines has the potential to contain domain shifts beyond the accepted causes in this research area (different part models, operating conditions and sensor locations) which can enable data leakage between training and test datasets. To demonstrate the impact of data leakage, six common bearing diagnosis methods are applied to two datasets using three data splitting methods to compare classification performance. Diagnosis is preformed using Principal Component Analysis (PCA), Supervised Principal Component Analysis (SPCA) and Linear Discriminant Analysis (LDA) in combination with frequency analysis and envelope analysis feature extraction methods. Datasets from McMaster University and Paderborn University are used as experimental data sources, and produce vastly differing results (over a 40% drop in accuracy) depending on the selected dataset splitting method, revealing a previously unknown domain shift. Despite great results for diagnosis methods using frequency response analysis on the data from McMaster, these results are not expected to generalize due to possible data leakage. Out of fifty-five previous works using the Paderborn dataset, ten are identified as likely to be affected and only six properly address the problem. Recommendations are given for future experiment design, model creation and model evaluation.

INDEX TERMS Bearing, fault diagnosis, vibration analysis, domain shift, data leakage, machine learning, train-test split.

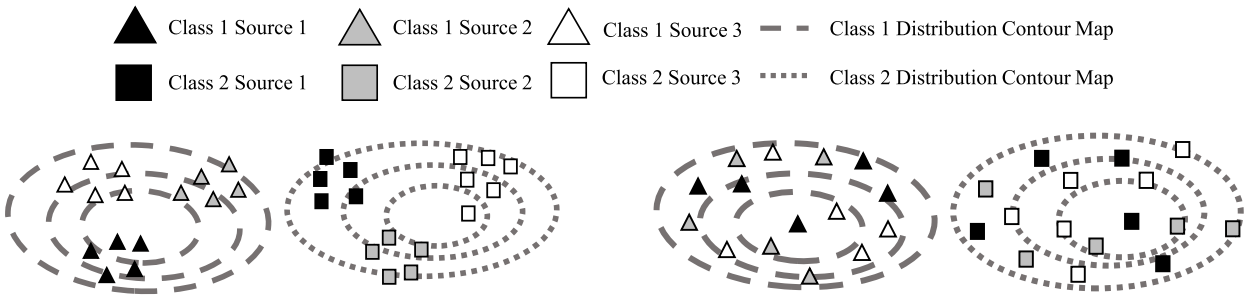
I. INTRODUCTION

Fault Detection and Diagnosis (FDD) is a research area that seeks to detect and identify faults, undesirable conditions, within mechanical systems before failure occurs, which could result in damage, expensive downtime or other consequences. Ideally, such conditions would be detected with sufficient time prior to failure so that corrective action can be taken. To accomplish this, FDD systems use a large variety of sensors, hardware, and software suited to the types of faults that they have been designed for. It is necessary that these FDD systems are reliable and robust, that their performance is consistent across the population of mechanical systems they will be expected to monitor.

In this context, FDD is considered as a problem where one diagnosis method can be used for an entire population of mechanical systems and it is simplified here to a classification problem (eg. healthy vs. fault condition identification). This population could be all motors of a single model, all parts of a particular model, a set of machines, or another mechanical system. For example, a quality system installed on an assembly line, state-of-health testing or a field testing system would fit this use case. These are applications where a healthy baseline of the system being monitored can *not* be established because historical data is not available. As such, these FDD systems should not be susceptible to small, acceptable

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(a) Samples drawn from different sources with a domain shift. Class distribution is not consistent from each source.

(b) Samples drawn from a class distribution without a domain shift. Class distribution is consistent across sources.

FIGURE 1. Examples of samples drawn from similar class distributions with and without domain shifts.

changes within the population of machines of interest, which can exist, in order to generalize to the entire population.

Machines that are identical on paper have slight differences, due to their own history and manufacturing tolerances, which have the *potential* to change measurements obtained from said machine. This has the potential to create a domain shift [1], a phenomenon that occurs when the probability distributions of the training and testing data differ, as illustrated in Figure 1. When domain shifts occur in data used for machine learning (ML), they can cause significant problems [1] and, therefore, negatively impact diagnosis methods using those ML techniques.

Certain sensors may be more likely to capture extra information unrelated to fault conditions than others. For example, the voltage at a single node may be more isolated from other elements of the system in a way that a microphone may not be. In other areas, experimental vibration data from rotating machinery has been found to contain machine-identifying information [2]. Vibration data may contain information specific to other components within the machine, other than the part of interest, even when those components are considered as ''healthy'' or ''undamaged''. This information may contribute creating unique vibration ''signatures'' that can be used to identify machines down to serial numbers [2]. These ''signatures'' are examples of a domain shift within the recorded vibration data.

In this paper, the impact of possible domain shifts are investigated for the problem of bearing fault diagnosis. In this context, the goal is to classify bearings based on their condition (healthy or a specific damage type/severity) using vibration data. Although other FDD applications also use vibration data [3], this particular problem is chosen for investigation because bearings are the leading cause of failure for motors [4], the common use of sensitive accelerometers for fault diagnosis [3] and vibration analysis is an important component to diagnosis methods [5], [6]. It is also an application where generalization across motors of the same model is expected and the application of model-free ML techniques is desirable, which tend to utilize more features, and thus may be at a higher risk of encountering a domain shift. In this context, *model-free* means there is no knowledge of the motor, parts or underlying system [2], but is not limited to, or implies, the use of neural networks.

As such the issue becomes not *if* there is a domain shift, but if there is a domain shift that can *impact* the diagnosis methods significantly. It's vital that the performance of our models is consistent, in the design phase and application phase, in order to select the best technique to tackle each problem. Over-inflated results in the design stage, which may be caused by over-fitting to small variations in data, are undesirable. The irrelevant differences within the physical system may not be large enough, or affect the physical phenomena being measured, to be captured by the measurement system, and therefore, the result is no *perceptible* domain shifts within the dataset. Alternatively, the domain shift may be present within the *raw data* but not the *feature(s) of interest* or after processing, at which point it is no longer a concern. Therefore, this includes several diagnosis techniques with a differing amount of frequency information, rather than selecting a single diagnosis method, in order to observe the impact on classification performance.

This paper does not address or investigate how possible domain shifts may be solved. Fortunately, there are many solutions available to do so, including methods developed for other types of bearing fault domain shift problems [7], [8] and general problems [9]. Domain shifting by itself is not a new concept in the field of bearing diagnosis as it is well-accepted that different operating conditions, machines/parts and sensors are all subject to domain shifts [3]. This work focuses on examining situations where those *accepted* domain shifts do not exist because unknown and unexpected domain shifts still cause difficulty for FDD techniques and are difficult to troubleshoot from model performance alone.

The contributions of this paper are to:

- Demonstrate the discrepancy in classification accuracy for ball bearing faults under different train/test splits, revealing the impact of data leakage through vibration data.
- Report the classification performance of established machine learning and feature extraction techniques on a new dataset.

- Examine the differing performance of two sensor placements.
- An overview of previous works in the area and their train/test splitting methods, to identify works which may be impacted by data leakage.

With this in mind, this paper will examine the impact of a domain shift within bearing fault vibration data by comparing the performance of six different model-free diagnosis methods with varying amounts of frequency information under different train/test splitting methods: *run-to-run*, *day-to-day* and *part-to-part*. The diagnosis methods tested are straightforward, utilizing standard feature extraction techniques, three common dimensionality reduction techniques (PCA, SPCA and LDA), and Support Vector Machines (SVM) for classification. The experiment is conducted on two datasets: Center for Mechatronics and Hybrid Technology (CMHT) Belt Starter Generator (BSG) Bearing Fault (BF) Dataset [10] from McMaster University and KAt Dataset [11], [12] from Paderborn University.

To summarize, this paper compares the performance of:

- Six bearing diagnosis methods using different common ML techniques on a closed dataset and an open datasets.
- Over different test/train dataset splitting strategies: *run-to-run*, *day-to-day* and *part-to-part*.
- Two tri-axial accelerometers, which are placed at different distances from the bearing of interest.

The remainder of this paper is organized as follows. A brief background of relevant literature including feature extraction and ML techniques. A thorough explanation of the diagnosis methods utilized. A detailed overview of the CMHT BSG BF Dataset (hereafter referred to as the CMHT Dataset), the experiment and results followed by the same for KAt Dataset. Followed by discussion of results for both datasets and concluding remarks.

A. BACKGROUND

Domain shifting is a well-known pattern recognition problem which may be the product of a change in feature(s) between sources and can lead to a change in class boundaries [1]. Domain shifts have the potential to pose issues for classification performance [1] and estimation of class probability distributions (see Figure 1) or measuring separability [67]. Domain shifts can also lead to problems evaluating the performance of models [68].

If the data contains additional information, and that data is used both for training and testing, then it can create a data leakage [68] problem, in that the model has been given access to data that it would not be expected to have during prediction. This can lead to over-optimistic evaluated model performance [68] and has the potential to impact common data splitting strategies used in bearing fault diagnosis [49], [63], where samples from the same signal can end up in the training and testing splits.

In this problem, data leakage and domain shifts are linked concepts. If the data collected from each source is the same (there is no domain shift) then there is no source-specific information available in the data. Thus, the sources can not be differentiated or re-identified (as all data is drawn from the same distribution) from the data itself. From the other side, any information about the data source would not be useful for classification could not generate data leakage.

In the opposite case, when the data sources are differentiable and linked to the class, data leakage can occur through the identification of the data source. In this way, the classifier can *bypass* the identification of the class from the data itself, instead it can learn to identify the source domain then associate that domain with a class label. Even if the data sources can not be identified with high accuracy, the information can be combined with other indicators to increase the overall classification accuracy for previously seen data sources. When this occurs, including data from the same sources in the training and testing sets leads to a case of data leakage, as the classifier would be expected to operate on unseen data sources. Therefore, the presence of a domain shift can enable data leakage.

1) DATASETS

Two existing datasets were used for this experiment, the first is the CMHT Dataset (Section III) is introduced in [10] and has previously been used in [69] and [70]. To check the results for consistency, a second experimental dataset was selected from existing benchmark datasets for validation. There are many open source experimental datasets for bearing faults available [3], [49] including the HUST Dataset [71] and KAt Dataset [11].

The KAt Dataset (Section IV) was chosen for this experiment because it contains vibration data and it contains the widest variety of individual bearings used for testing (thirty-two different parts total) [11]. As discussed later in Section IV, the KAt Dataset contains examples of six healthy parts of the same model of bearing, whereas most other datasets had a very limited number of healthy examples. This dataset is also the closest match to the CMHT Dataset in terms of conditions and types of data collected.

With the KAt Dataset, [11] uses several parts entirely for testing (fault type as class label) but a wide variety of splitting strategies have been used. On the same data, [49] uses data from all selected parts in training and testing (parts as class labels) using order and random data splitting while [63] uses only random. The lack of consistency for data splitting and model evaluation, even within the same dataset, makes comparison across different papers difficult, but also introduces confusion as to which split method ought to be used.

A review of FDD techniques that have been evaluated using the KAt dataset was conducted to identify the most common train/test splitting methods. The results and their categories are summarized in Table 1, focusing only on FDD methods that made use of the vibration data of the KAt dataset over the last five years and excluding transfer learning

TABLE 1. Splitting methods used. Sample selection methods are enclosed in square brackets.

Split		Validation		
		KAt Only	KAt + Other Benchmark	KAt + Experimental (Optional: + Benchmark)
Separate Parts	Set: Artificial/Real	[11], [13], [14]		
	Set: Real Damage	[11], [14], [15]		
	Set: Multiple Damages	[11]		
	Set: Custom	[16]	[17]	
Unknown	Set: Real Damages [Not Reported]	[18], [19]	[20], [21], [22]	[23]
	Set: Real Damages [Unclear]	[24]	[25], [26]	[27]
	Set: Not Reported/Unclear	[28]	[29], [30], [31], [32]	
	Set: Custom	[33]	[34], [35], [36], [37], [38], [39]	
Mixed Parts	Set: Real Damages [Random]		[40]	
	Set: All [Random]	[41], [42], [43]	[44]	
	Set: Custom [Random]	[42], [45], [46], [47], [48]	[49], [50], [51], [52], [53]	[54]
	Set: Not Reported/Unclear	[55]		[56], [57]
	Set: Custom [Unclear]	[58], [59]	[60], [61], [62], [63], [64]	[65]
	Set: Various [Order Split]		[49]	[66]

problems. Fifty-five works are included in this review, of those, twenty-two did not provide enough information to determine if data from any parts existed in both training in testing datasets. Of the remaining thirty-two works, only six took steps to ensure that part data was not mixed between the sets. Model performance from these papers are collected for comparison purposes in Tables 8 and 7.

Based on this information, it was decided that two different train/test splits would be used on each dataset, to check if there would be any significant difference in model performance. A similar test was conducted by [49], but the difference in model performance was only briefly touched on and identified as an interpretability issue (a model problem), rather than a domain gap (a data problem). Since [49], there has been a lack of further investigation into this topic. In contrast to [49], this experiment is preformed with a larger subset of the dataset, less complex classification techniques as used and two datasets are used.

2) PREVIOUS TECHNIQUES

There is a large number of model-free ML techniques that have been developed to solve the bearing fault diagnosis problem [49]. The methods used in this paper were selected because of popularity, presence in existing literature and simplicity. Many different frequency extraction techniques have been used for rolling bearing fault detection and diagnosis including: Fast Fourier Transform (FFT) [30], [49], Short Time Fourier Transform (STFT) [49], wavelets [49], envelope analysis [5], [6], [30] and more [49]. Frequency analysis and envelope analysis are well-known techniques for tackling this problem and will be used in this paper.

PCA, SPCA and LDA are used as dimensionality reduction techniques (DRT), because they are common, explainable and are all linear transformations. Additionally, their computation time is comparatively low relative to other DRT, reducing training time and allowing for many tests to be run in a very short time. Existing literature for bearing faults has used PCA [5], [52], [67], [72], [73] and LDA [52], [67]. As PCA is susceptible to class imbalance, SPCA is also included. These DRT are combined with SVM, a popular classifier for solving complex, small sample pattern recognition problems with high accuracy and good generalization [74]. SVM has also been used in other bearing fault diagnosis methods [11], [75].

PCA is a commonly used technique for dimensionality reduction, noise reduction [76] and data visualization that captures the vectors with the highest variances into principal components [74]. SPCA [77], [78] is based on weighted PCA where the weights are determined using the class labels.

LDA can be used for supervised multiclass dimensionality reduction [74]. In this case, the dimensions are the vectors of highest class separability under the assumption that the classes have Gaussian distributions and the same covariance matrix. In the case of a domain shift, there will be a difference in the probability distribution of a class, which can change the distribution to be multi-modal and therefore classes would *not* have a Gaussian distribution. However, in the case that there is no domain shift, the only source of noise should be the measurement noise, which is expected to be Gaussian.

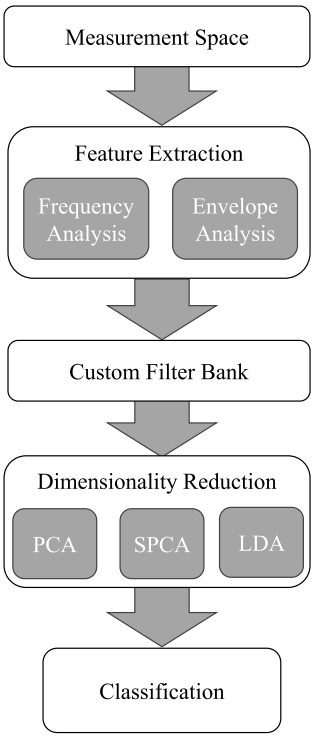


FIGURE 2. Applied machine learning (ML) pipelines for fault diagnosis.

II. APPLIED DIAGNOSIS METHODOLOGIES

This section covers the technical implementation of the diagnosis pipelines tested (as illustrated in Figure 2), in which two frequency extraction techniques and three DRT are used. Parameters mentioned in this section are used on both dataset unless stated otherwise (Sections III and IV). In every case, the method is model-free and has no additional input as to the operating conditions. Thus, the model is only trained and applied to one operating condition at a time.

The process is as follows:

Step 1. Measurement space/Raw Data: Collect the vibration signals under steady state conditions.
- (i) Every file is pre-processed with a low pass Butterworth zero phase digital filter [79].
- (ii) All data is resampled to 48 kHz.

Step 2. Feature Extraction: For each vibration signal, construct a set of segments of window size w and incrementing by k samples.
- (i) Frequency Analysis: Extract power spectrum using FFT [80] scaled by a Blackman window.
- (ii) Envelope Analysis [71]: The signal is scaled to zero mean, and the Hilbert transform [79] is applied to get the analytic signal. Then, the absolute value of the analytic signal is calculated then the envelope spectrum is obtained using FFT scaled by a Blackman window.

For multiple channels, the magnitude over all channels at each frequency is used. Values are converted to decibels (dB).

Step 3. Custom Filter Bank: Remove value for 0 Hz and values for all frequencies above cutoff parameter f_c.

Step 4. Dimensionality Reduction: Scale all remaining features to have zero mean and unit variance then apply PCA, SPCA, or LDA.

Step 5. Classification: Scale each feature linearly to -0.5 to 0.5 (hard limit of -1 to 1) then apply multi-class SVM.

Note that Steps 1-3 would typically all be considered to fall under the umbrella of data pre-processing, but they have been split out into different steps in this case for clarity. Further details on each step are included below, and additional details for each dataset are listed in their respective sections.

(Step 2) A sliding window is applied to create a set of samples for each file, in this case, a window size of 8192 (171 ms) is used with an offset of 2048. Feature extraction is then applied to each of these segments separately.

The window size is chosen to be large enough to adequately capture the harmonics of the slowest operating speeds from both datasets, which in this case, is 900 RPM (see Sections III and IV for further details). The same window size and offset is used for every operating speed to maintain consistency. Train/test data splitting is covered in Sections III-A and IV.

(Step 3) The purpose of the filter bank is to control the size of the feature space used for ML by varying the value of f_c, as the lower bound of the custom filter bank is never changed. As the value of f_c increases, more features are passed to the DRT. Consequently, the scale of the dimensionality reduction problem and the chance of feature(s) subjected to a distribution shift being included in the data increases.

(Step 4) For the dimensionality reduction stage, PCA [81] and LDA [81] use default options[1] unless otherwise stated. For SPCA [78], the following parameters are used: attraction $\alpha = 10$ and repulsion $\beta = 1$. The purpose of using a very large weight for α is to significantly differ the results of SPCA from PCA (as shown in Figure 5). With these parameters, SPCA becomes a nice middle ground between PCA and LDA.

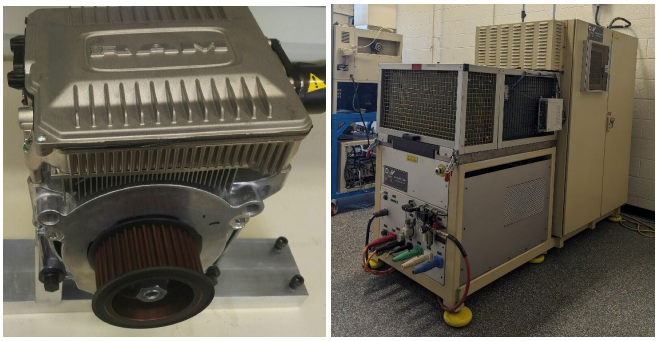
(Step 5) After dimensionality reduction, data is linearly scaled before being passed to a SVM [81], which is used for classification in each case to maintain consistency, with default parameters[1] other than $C = 0.2$ and class_weight is balanced. In this case, the regularization parameter was manually selected based on a small subset of the CMHT Dataset.

It should be noted these selected parameters may not be optimal, for the problem, the datasets themselves, or the individual train-test instances. Optimization of these parameters is possible, however, it is not necessary for the purpose of this study. To limit possible confounding factors, the use of consistent parameters is more important.

III. APPLICATION: CMHT DATASET

Bearing fault vibration data from a Belt-Starter Generator (BSG) electric motor (FCA US LLC) using a testbench

[1] Scikit-learn 1.4.1 Documentation: PCA, LDA and SVM.

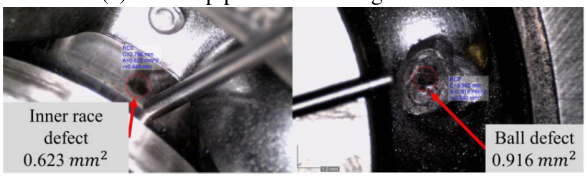


(a) BSG (left), and HT-250 test bench (right).

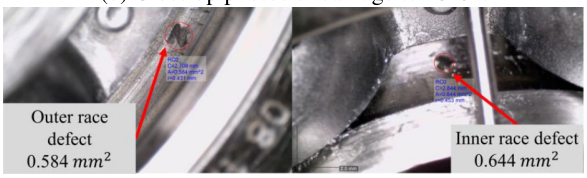
FIGURE 3. CMHT BSG test setup [10], [69].



(a) Close up photos of damage to F3-01.



(b) Close up photos of damage to F5-01.



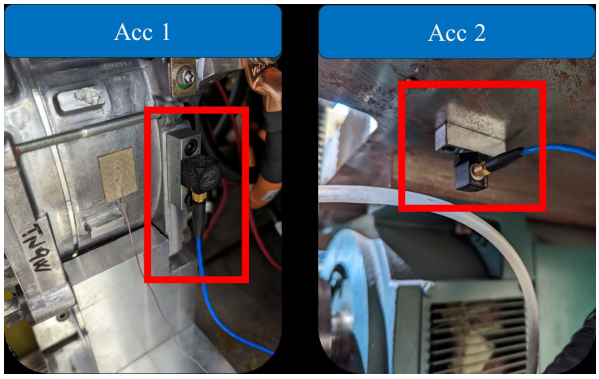
(c) Close up photos of damage to F7-01.

FIGURE 4. CMHT BSG bearing faults [10], [69].

(Model HT-250, D&V Electronics LTD.) was obtained from CMHT [10]. The test setup is shown in Figure 3 and the bearings are shown in Figure 4. More detailed information about the dataset is provided in [69] and [70]. The dataset includes several combinations of artificial fault conditions for the drive-end bearing over twenty different speed and load conditions from 1000 to 17000 RPM. In this case, a subset of the dataset is used (shown in Table 2), which excludes all lubrication fault conditions.

Vibration data from two tri-axial accelerometers (Model No. 356A16, PCB Piezotronics, Inc.) were used, as shown in Figure 3b, and all channels were sampled at 48 kHz [10], [69]. The ±10% frequency range of the accelerometer is 0.3 to 6000 Hz [82]. The dataset also contains audio, current and voltage data but only vibration was used.

There are four classes used for this experiment and details of the fault conditions are described in Table 2.



(b) Location of accelerometers on motor.
Accelerometer 1 (A_1): Right side of the BSG's chassis.
Accelerometer 2 (A_2): Underneath the test bench.

TABLE 2. Experimental data used from the CMHT Dataset [10], [70]. There is the same number of runs for each operating condition. The area of the respective hole is listed next the defect type. Note: day identifiers are not in chronological order.

State	Bearing ID	Day ID	Number of Runs
Healthy	AM-01	Day 1	2
		Day 2	2
		Day 3	1
		Day 4	1
	AM-02	Day 5	2
	AM-03	Day 6	2
	OEM-01	Day 7	1
		Day 8	1
	OEM-02	Day 9	1
		Day 10	2
		Day 11	2
		Day 12	2
		Day 13	1
OR (1.033 mm^2) BD (0.957 mm^2)	F3-01	Day 14	2
		Day 15	2
IR (0.623 mm^2) BD (0.916 mm^2)	F5-01	Day 16	2
		Day 17	2
IR (0.584 mm^2) OR (0.644 mm^2)	F7-01	Day 18	2
		Day 19	2

OEM: Original Equipment Manufacturer;
AM: Aftermarket; BD: Ball Defect;
OR: Outer Ring Defect; IR: Inner Ring Defect;

Fault conditions were artificially created in aftermarket (AM) bearings using a drill [10], [69]. The AM bearing model number is SKF 6305-2RS1/C3 and the Original Equipment Manufacturer (OEM) bearing is a custom modified SKF bearing with model number: SKF BB1-0890BA [10], [69]. There is a significant class unbalance within this dataset (see Table 2) of healthy to faulty examples. To account for this, the error was averaged so each part (Bearing ID) has equal

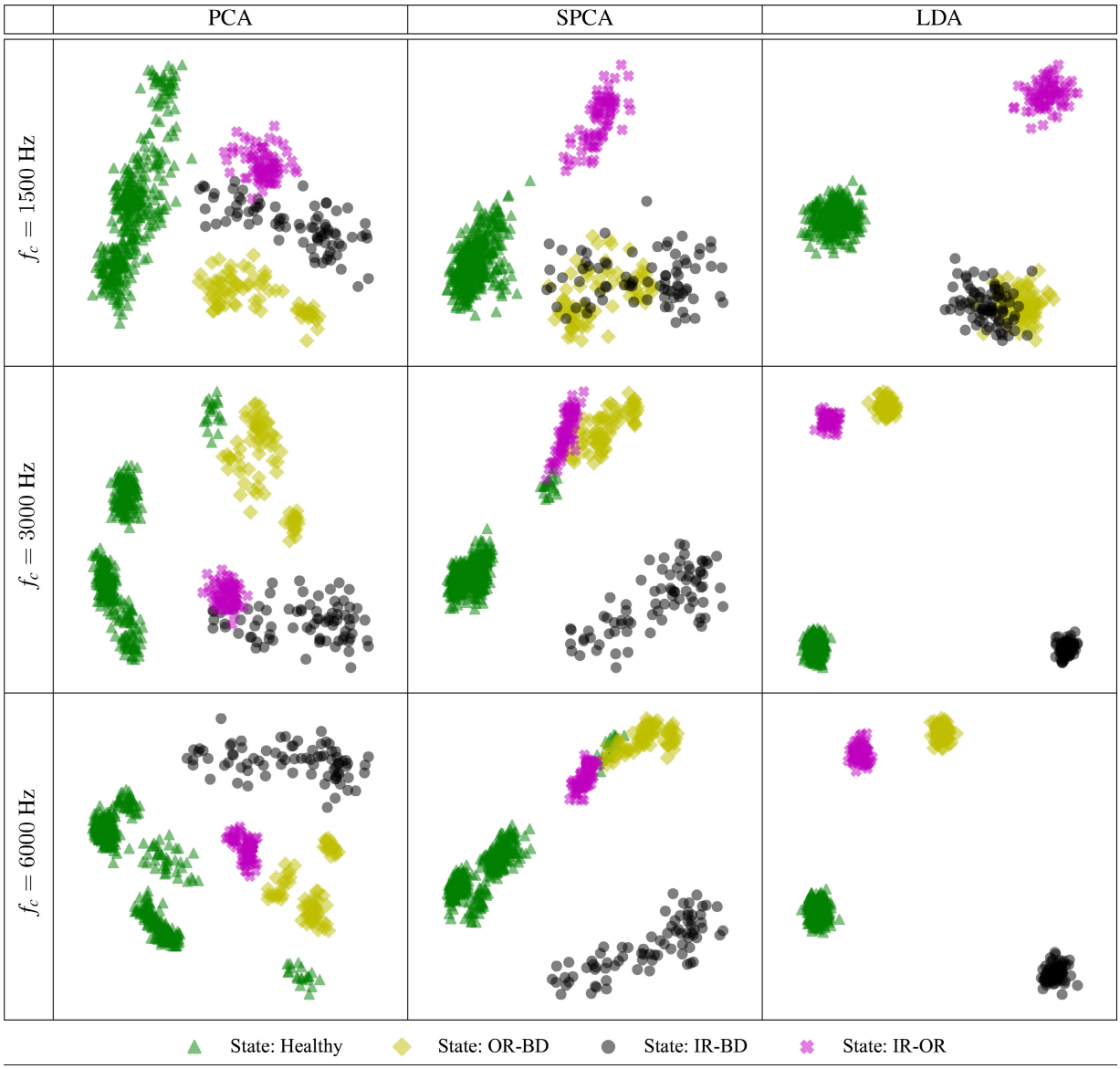


FIGURE 5. Example 2D plots of DRT outputs for different methods and frequency cutoffs with the CMHT Dataset. On each plot, the X axis is first component and the Y axis is second component. Created using data from A_1 for one operating condition with frequency analysis.

weight within it's respective class and each class has equal weight in the overall error.

The number of *runs* used for the experiment are listed in Table 2. In this paper, one *run* is one continuous period of data collection at steady state during a *single test* of the machinery. Each run contains approximately one second of data and is stored as a single file. The time difference between runs on the same day is a minimum of two hours rest. The rest time between days is more variable, ranging from the next day to several weeks.

Within this dataset, there are some known sources of external disturbances. All data was collected on BSG-1 except for bearing OEM-02, which was collected with BSG-2, a second motor of the exact same model [10]. F3-01, F5-01, AM-02 and AM-01 (Days 3 and 4) were collected on BSG-1 with a new rotor after a required repair [70]. No data with rotor damage is included. Initial temperature varied between 18-21°C and peak temperature was between 55-65°C [10], [69]. In addition, the act of swapping out the bearing may itself introduce some variation as some disassembly is required.

A. EXPERIMENT SETUP

To observe the possible impact of a domain shift between data collected on different days, depending on the amount of information used to construct the diagnosis model, the classification performance for each pipeline under different testing methods is observed. The following variation of leave-one-out (LOO) dataset splitting are used:

- In *run-to-run*, all samples from a single run are used for testing. There are 32 runs in total.

TABLE 3. Highest accuracy for each method for the CMHT dataset in percent. Values over ninety-five percent are underlined.

Method		A_1		A_2	
		Run-to-Run	Day-to-Day	Run-to-Run	Day-to-Day
Frequency Analysis	PCA	96.73	94.73	88.58	84.53
	SPCA	98.28	96.18	96.95	91.50
	LDA	98.89	97.78	95.69	90.52
Envelope Analysis	PCA	88.30	86.08	83.69	79.18
	SPCA	95.16	90.96	92.16	84.94
	LDA	95.92	91.18	89.54	80.54

- In *day-to-day*, all samples from a single day are used for testing, which may contain data from one or two runs. There are 19 days in total.

Data reserved for testing is not used for any portion of training. For *day-to-day*, days that have a single run are included. The reduction in the number of samples for training from *day-to-day* (up to 6% of dataset for testing) to *run-to-run* (3% of dataset for testing) is not expected to have a significant impact on classification performance. LOO is used as the basis because it is an easily reproducible dataset split method and maximizes the amount of data available for training.

The state is used as the class label (Table 2). As there are four classes, the maximum number of dimensions obtained from LDA is three [74], therefore, the number of dimensions preserved using PCA and SPCA is also set to three. The pre-processing filtering cutoff is 10 kHz and f_c is tested from 500 Hz to 6000 Hz at 500 Hz increments.

B. RESULTS

Example graphs for two possible values of f_c are shown in Figure 5. At $f_c = 6000$ Hz, PCA produces clear sub-clusters within the Healthy and OR-BD classes while LDA produces tightly grouped separable single-class clusters. Recall that the Healthy class contains multiple parts and a second motor, which are likely contributing factors to the multiple sub-clusters within that class. In this plot, PCA is suggesting there may be a domain shift in the original data. However, with the other methods and PCA at $f_c = 1500$ Hz, these sub-clusters are not as visible or entirely indistinguishable.

Figures 6 and 7 show the classification performance over all tested values of f_c for both accelerometers and feature extraction methods. With A_1 using frequency analysis, all methods produce at most a 0.04 error rate with the maximum number of features, a very good result. Comparing results between frequency analysis and envelope analysis, frequency analysis consistently results in lower error, although the difference is minor in some cases. Out of all tested DRT, SPCA produces the lowest error most consistently and levels off at around 3 kHz.

There is a large visible "bump" in the error rate for LDA for f_c between 2.5 kHz to 4 kHz for both feature extraction methods and also across both accelerometers. This behavior may be due to the introduction of features affected by a domain shift near those frequencies, which would produce a non-Gaussian distribution. Note that even though the measurement noise distribution is Gaussian, the noise distribution of the individual features is skewed due to the conversion to dB (Section II). As LDA assumes a Gaussian class distribution, these factors may affect the DRT and cause it to behave unexpectedly.

After this "bump", LDA produces the lowest error at 6 kHz, the limit of f_c. In the case of A_1, using an f_c value above 3 kHz appears to have limited benefits for the performance of PCA and SPCA, as the error rate curve levels off. This is not true for A_2, where including the higher frequency data does slightly increase performance.

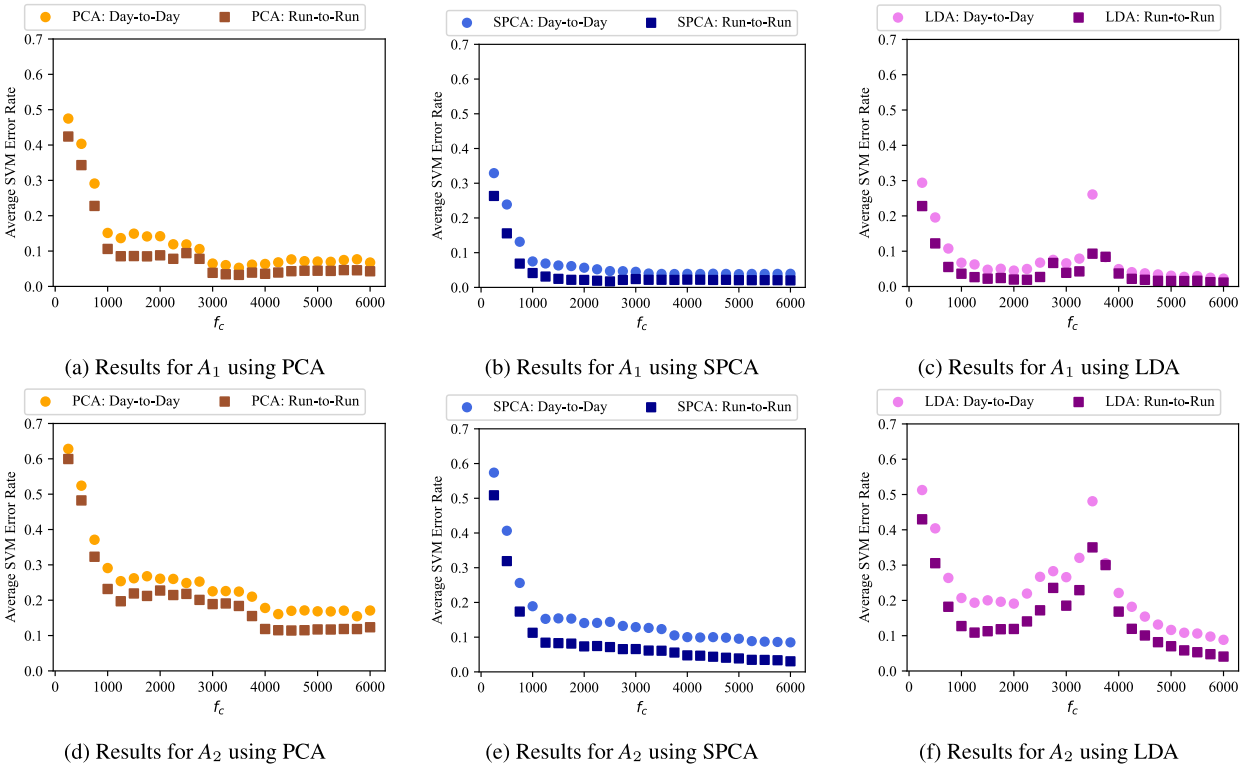
Comparing the two accelerometer locations, A_1 produces better results than A_2, which was expected as A_1 has a better signal-to-noise ratio (SNR). A_2, despite being further from the bearing, is still useful for classification, but also tends to display a slightly greater difference in *run-to-run* vs. *day-to-day* performance. From the *run-to-run* split results for frequency analysis for A_1 and A_2 (Figure 6), both sensors pick up enough information to achieve near average zero error with SPCA, given the maximum number of features.

From the results on the CMHT Dataset (Figures 6 and 7), the difference between *run-to-run* and *day-to-day* can be observed, and in every case, *run-to-run* has a lower error than *day-to-day*. For each DRT, the maximum difference in error rate between the splitting methods varies, with PCA having a difference of up to 0.07, SPCA up to 0.09 and LDA up to 0.17. Table 3 shows the differences in maximum accuracy. From this, it can be seen how having run information from the same day available to the model is aiding classification. Therefore, there is day-to-day variation within the dataset that is resulting in an unexpectedly lower error rate for *run-to-run* train/test splitting.

IV. APPLICATION: KAT DATASET

The KAt Dataset [11], [12] is provided by Paderborn University and contains examples of healthy bearings, artificially damaged bearings and real damaged bearings. The sampling rate of the original accelerometer (A_3) is 64 kHz, which is resampled to 48 kHz, to enable the use of same parameters as the CMHT Dataset. The dataset contains multiple examples of 6203 bearings under four operating conditions. As with the CMHT Dataset, all four operating conditions are trained and tested independently then the average error is reported.

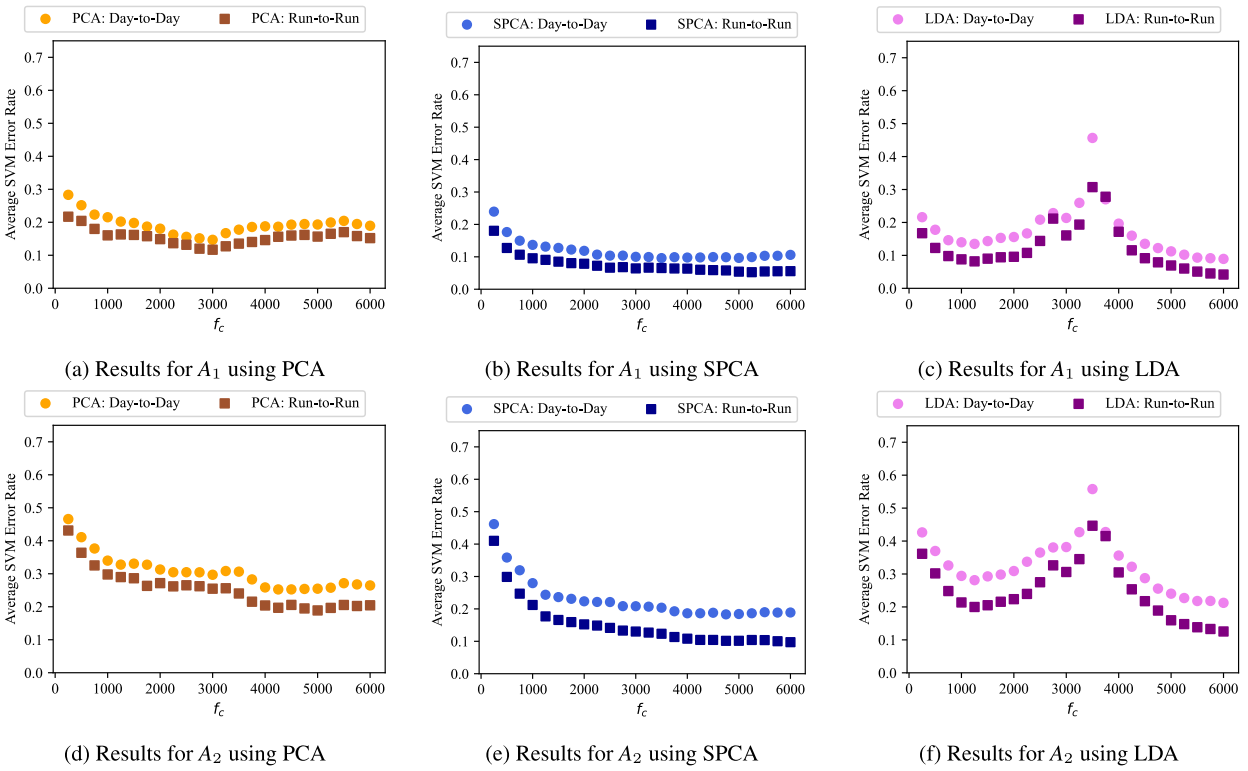
A subset of the above dataset is used for this experiment, as listed in Table 4, with state used as the class label. Only



(a) Results for A_1 using PCA (b) Results for A_1 using SPCA (c) Results for A_1 using LDA

(d) Results for A_2 using PCA (e) Results for A_2 using SPCA (f) Results for A_2 using LDA

FIGURE 6. CMHT Dataset results for frequency analysis. Operating frequency range: 17-283 Hz.



(a) Results for A_1 using PCA (b) Results for A_1 using SPCA (c) Results for A_1 using LDA

(d) Results for A_2 using PCA (e) Results for A_2 using SPCA (f) Results for A_2 using LDA

FIGURE 7. CMHT Dataset results for envelope analysis. Operating frequency range: 17-283 Hz.

healthy bearings and artificial fault conditions that have more than one bearing listed for that damage combination are included (insuring there is at least one part with that damage type used for training). As with the CMHT Dataset, the error is weighted to account for class imbalance.

The following ways of splitting the dataset are used:

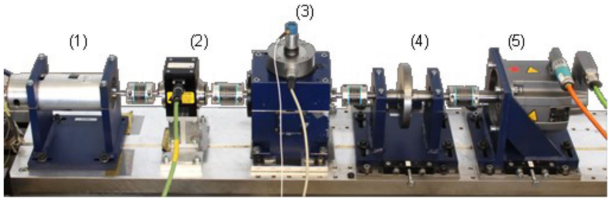


FIGURE 8. Modular test rig used for KAt Dataset data generation [11]. (1) is an electric motor. (2) is a torque-measurement shaft. (3) is a rolling bearing test module. (4) is a flywheel. (5) is a load motor.

TABLE 4. Subset of parts used from KAt dataset [11]. OR: Outer ring defect. IR: Inner ring defect.

State	Damage Type	Severity	Bearing Code
Healthy	N/A	N/A	K001, K002, K003, K004, K005, K006
OR	Electric Engraver	2	KA03, KA06
	Drilling	2	KA08, KA09
IR	Electric Engraver	1	KI03, KI05
	Electric Engraver	2	KI07, KI08

- In *run-to-run*, all samples from a single file are used for testing. There are 280 runs in total.
- In *part-to-part*, all samples from a single part are used for testing, as identified by Bearing Code (see Table 4). There are 14 parts in total.

Examples of each case are given in Tables 5 and 6. Although not listed, all files would use the same operating condition prefix (such that operating conditions are not mixed).

The same parameters from the CMHT Dataset are used, with the following exceptions: LDA produces two features instead of three, f_c is from 500 Hz to 12000 Hz (the 10% range of the accelerometer [83]) with 500 Hz increments and the pre-processing filtering cutoff is 20 kHz.

A. RESULTS

From the KAt Dataset results in Figures 9 and 10, the difference between *run-to-run* and *part-to-part* can be observed, which is much larger than the *run-to-run* and *day-to-day* difference in the CMHT Dataset. *Run-to-run* always achieves a much lower error rate than *part-to-part*, indicating the very large benefit of including data from the same part in training data.

For all methods, there is a huge jump in classifier error rate for *part-to-part* vs. *run-to-run* (up to 0.47). The overall performance on *part-to-part* is extremely inaccurate, reaching a minimum error of 0.39. Note that a classifier which generates randomly predicted class labels would result in an error rate of 0.67, therefore SPCA does only slightly better than guessing on *part-to-part*. PCA has the best *part-to-part* average error rate across all values of f_c at 0.44, however, this is still a poor result. For PCA and LDA, over 4 kHz has minimal benefit but for SPCA, error decreases until about 8 kHz.

TABLE 5. Example of train and testing sets for ***Run-to-Run*** dataset splitting. X denotes all files with that prefix (numbers 1 to 20).

KAt Split Type: *Run-to-Run*		
Train Set	Test Set	Task Number
K001_2 to K001_20, K002_X, K003_X, K004_X, K005_X, K006_X, KA03_X, KA06_X, KA08_X, KA09_X, KI03_X, KI05_X, KI07_X, KI08_X	K001_1	1
⋮	⋮	⋮
K001_X, K002_X, K003_X, K004_X, K005_X, K006_X, KA03_X, KA06_X, KA08_X, KA09_X, KI03_X, KI05_X, KI07_X, KI08_1 to KI08_19	KI08_20	280

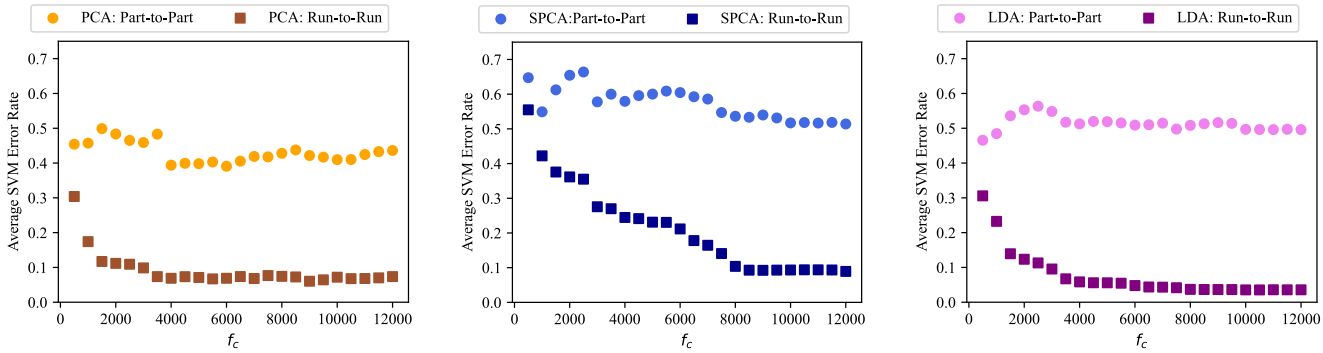
TABLE 6. Example of train and testing sets for ***Part-to-Part*** dataset splitting. X denotes all files with that prefix (numbers 1 to 20).

KAt Split Type: *Part-to-Part*		
Train Set	Test Set	Task Number
K002_X, K003_X, K004_X, K005_X, K006_X, KA03_X, KA06_X, KA08_X, KA09_X, KI03_X, KI05_X, KI07_X, KI08_X	K001_X	1
⋮	⋮	⋮
K001_X, K002_X, K003_X, K004_X, K005_X, K006_X, KA03_X, KA06_X, KA08_X, KA09_X, KI03_X, KI05_X, KI07_X	KI08_X	14

Maximum accuracy for each method is listed for comparison in Table 8. Tables 8 and 7 provide reported accuracy values of previous works. Note that none of the other reported values use the exact same split as this experiment, so they are not a valid direct comparison.

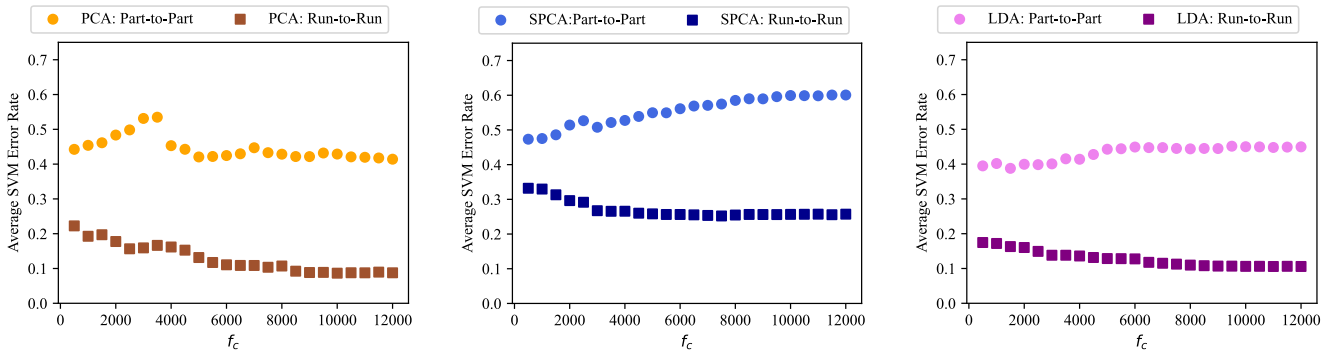
V. DISCUSSION

When only considering the *run-to-run* performance (Figure 6 vs. Figure 9 and Figure 7 vs. Figure 10), there are large differences between the two datasets. Specifically, the large difference between *part-to-part* versus *run-to-run* accuracy in KAt and the much smaller difference between *day-to-day* versus *run-to-run* in the CMHT dataset. There are several possible causes: the test setup, the sensors used for data collection, the known differences in the CMHT Dataset, and the types of fault conditions. Firstly, the two datasets used



(a) Results using PCA (b) Results using SPCA (c) Results using LDA

FIGURE 9. KAt results using frequency analysis. Operating frequency range: 15-25 Hz.



(a) Results using PCA (b) Results using SPCA (c) Results using LDA

FIGURE 10. KAt results using envelope analysis. Operating frequency range: 15-25 Hz.

different motors under different operating conditions, and there is the possibility that this influences the strength of the signals.

Using tri-axial accelerometers may also be a factor, as they may be better at capturing fault signatures, but also increase the amount of irrelevant information. In theory, A_1 has the best placement, as it is closest to the bearing, but that does not directly translate to the lowest *run-to-run* error. A_2 sensor placement is far from ideal, however, it does still get signals from the machinery under test, and can produce similar results to A_1. Placement close to the bearing is known to be important [5], however less than ideal placement options may still show some promise.

In addition, the combination of two fault conditions at once should produce a fault signature that would appear in more frequency bins. In theory, this could create an easier classification problem. Although, fault severity within the CMHT Dataset is lower than the KAt Dataset for most parts (see Tables 2 and 4).

The additional known differences within the CMHT Dataset (Section III) create a larger domain shift than is present in the KAt Dataset, enabling more data leakage on *run-to-run*, and therefore inflating performance. The CMHT test bench is not modular, unlike the KAt one, and the bearing swap process required several steps and partial disassembly. This may also have contributed to some of the differences within the CMHT Dataset, exacerbating any domain shift caused by a part swap. However, this swapping process is also more realistic for motors in use, as the process used would be the same for a repair and the diagnosis system would be expected to perform the same in such a scenario.

A. PERFORMANCE GAPS

Comparing the results from Sections III-B and IV-A, *run-to-run*, *day-to-day* and *part-to-part* all have differences in error. If the classification error was caused by a high class mixture (ex. measurement noise), then each dataset split method should all have approximately the same error rate. Therefore, there are domain shifts that allow for data leakage, which can influence the error rate up to a large amount (up to a difference of 0.47).

From the KAt Dataset results (Section IV-A), there is a large performance difference from *run-to-run* to *part-to-part*, which suggests that there may be data leakage for specific *parts* that can have a large influence on classification results. The results on the CMHT Dataset from *day-to-day* splitting

TABLE 7. Top reported accuracies and methods for KAt dataset when parts are mixed. Methods that are tested on more than one split are in bold. Values over ninety-five are underlined.

Split/Subset		Method	Highest Accuracy
Mixed Runs	Real Damages (Random)	SVM	85.75 [40]
		ECNN	98.17 [40]
	All (Random)	CSSL	96.35 [44]
		DRN-STFT	99.86 [42]
		ATVMF-RF-Cost-ACO	93 [41]
		VI-GCN	93.74 [43]
	Custom (Random)	**FFT-CNN-DA (Best)**	96.62 [49]
		FFT-AE-DA (Best)	98.62 [49]
		ISCNN	93.9 [50]
		CNN	99.7 [46]
		DNN	95.07 [47]
		FAE	99.81 [53]
		DRSN-BiLSTM	92.69 [51]
		TET-LDA-NNC	99.08 [52]
	Various (Order)	**FFT-CNN-DA (Best)**	76.92 [49]
		FFT-AE-DA (Best)	99.54 [49]
		SBC-TFW	99.88 [66]

TABLE 8. Top reported accuracies and methods for KAt dataset for this experiment and reported results for part splits from literature. Methods that are tested on more than one split are in bold. Values over ninety-five are underlined.

Split/Subset		Method	Highest Accuracy
Part Split	Artificial/Real	**Ensemble**	75.0 [11]
		CART	65.9 [11]
		RF	64.1 [11]
		SVM-PSO	65.5 [11]
		DAT	77.64 [13]
		MMD-DAT (max)	90.86 [13]
		CNN	80.59 [14]
		CDCN	79.2 [14]
		MABFN	89.53 [14]
	Real Damages	**Ensemble**	98.3 [11]
		CART	98.3 [11]
		RF	98.3 [11]
		SVM-PSO	75.8 [11]
		CNN	88 [15]
		CNN	95.77 [14]
		CDCN	79.2 [14]
		MABFN	99.86 [14]
	Multiple Damages	**Ensemble**	91.3 [11]
		CART	91.3 [11]
		RF	91.3 [11]
		SVM-PSO	70.8 [11]
	Custom	Multi-layer SNN	75.97 [17]
		ACDIN	94.50 [16]
		SVM	70.45 [16]
	Part-to-Part	**SVM-PCA-FFT**	60.92
		SVM-SPCA-FFT	48.59
		SVM-LDA-FFT	53.44
		SVM-PCA-EA	58.58
		SVM-SPCA-EA	52.67
		SVM-LDA-EA	61.21
Mixed Parts	Run-to-Run	**SVM-PCA-FFT**	93.97
		SVM-SPCA-FFT	91.08
		SVM-LDA-FFT	96.46
		SVM-PCA-EA	91.35
		SVM-SPCA-EA	74.79
		SVM-LDA-EA	89.42

may suffer from data leakage as well due to data from the same parts often being present within the training and testing datasets. The lack of additional parts with fault conditions within the CMHT Dataset prevents the conclusion that the model has successfully learned to identify fault conditions. At the time, the differences in data from sources other than the fault conditions was expected to be minimal based on existing literature in the area of bearing fault FDD and domain shifts were not expected.

This begs the questions if the parts themselves can be re-identified by classification. Since [42], [50], [52], [54], [58], [61], [63], [64] all use Bearing IDs from the KAt Dataset as class labels, with high success rates, they had unintentionally demonstrated how machine learning techniques can be very accurate for part re-identification (even under different conditions). Therefore, the re-identification of specific bearings with vibration data is very possible.

Based on this information, data collected from the same bearing correlates together too strongly to be considered as independent samples and allows for data leakage. Therefore, the iid (independent and identically distributed) assumption required for many machine learning techniques does not hold true. In this case, the problem is being examined from the standpoint of model *evaluation* rather than model *training* but this may also pose issues during training, for example, it may encourage over-fitting.

Note that for this problem (defined in Section I), it would be expected for a FDD model to not have observed the part being tested. *Run-to-run* and *day-to-day* splitting therefore gives the model information it should *not* have. These splits are examples of data leakage, causing the model to produce overoptimistic results in testing and it's performance is not expected to generalize [68].

However, *run-to-run* may still be a useful train/test split for other purposes than model evaluation. Troubleshooting, consistent reporting, and comparisons are examples of such cases. It would be expected that diagnosis methods should have consistent performance on new data obtained from the

TABLE 9. Previous works by publication year and level of part mixture.

Publication Year	Mixed Parts		Unknown		Separate Parts
	KAt Only	KAt + Other	KAt Only	KAt + Other	
2024		[44] [57], [64], [65], [54]	[24]	[20], [25], [38]	
2023	[41], [48], [55], [58]	[56], [60], [66]	[28]	[23], [29], [30], [31], [35], [36]	[14]
2022	[42], [45], [46], [47]	[50], [51], [61], [62], [63]		[27], [32], [37]	[17]
2021	[43]	[52]	[33]	[21], [22], [34], [39]	[13]
2020		[53], [49]	[18], [19]	[26]	[15]
< 2020	[59]	[40]			[11], [16]
Count	10	17	5	17	6

same part collected under the same conditions. Results can be compared against other train/test split methods to check the impact of data leakage on performance.

B. ANALYSIS OF METHODS

As this study is limited to vibration data, domain shifts in other measurements used for bearing fault diagnosis have not been examined. It is possible this phenomenon does not occur as severely in other sensors used for bearing fault diagnosis, such as current and voltage measurements. In this application, vibration data may *appear* to be a better data source for the prediction of fault conditions than other types due it's performance being inflated by data leakage, rather than the sensor's ability to capture fault signatures. Therefore, when preforming cross sensor comparisons with model-free ML techniques, possible data leakage for each type of sensor should be taken into account.

In most cases, higher values of f_c decrease the error rate, however, this is not always the case. As mentioned in Section III-B, use of LDA showed a large increase in error as some features were introduced. Other methods also experienced smaller bumps or increases (Section IV-A). As more features are passed to the DRT, the difficulty of the ML task is increased, in addition to the number of domain shifted features that may be introduced. It is important to recognize that blindly including additional features may not always be beneficial.

On the topic of the performance of the diagnosis methods themselves, the only test for generalization is on the KAt Dataset *part-to-part* results, in which performance is very poor in all cases. This is not to say that PCA, SPCA, or LDA are not useful as part of diagnosis methods or should not be used, but they may not preform as expected when there are domain shifts within the data. Nor may the impact of any domain shifts be limited to these techniques, other model-free ML techniques may also be susceptible. Model-based techniques may be preferable to model-free techniques for this reason.

To approach this problem with model-free techniques, the domain shift must be addressed to ensure consistent performance. One possible solution may be to find a way to eliminate the domain shift from the data, such as through domain adaptation. In the case that the domain shift is limited to specific features that are not useful for classification, then the problem is to simply detect and remove features that show a domain shift, through feature selection/filtering. When domain shifts exist in features that are also needed for classification, the problem becomes more complex and requires domain adaptation. Determining which of these cases exist in this application is beyond the scope of this paper, and the use of additional techniques does not remove the need to validate models against new data sources.

C. IMPACT

Although bearing FDD is not usually listed as an area of concern for data leakage in machine learning applications [68], that does not mean the issue does not exist. Despite the original dataset paper for the KAt dataset [11] using part by part train/test splits, recent work continues to use mixed part train/test sets (as shown in Table 9). While this may be required for the application of some machine learning techniques, it also makes them susceptible to data leakage.

Table 9 identifies the works that use the KAt dataset which are most at risk of results being influenced by data leakage. The ten works listed under "Mixed Parts: KAt Only" are the most uncertain because their splitting methods might the reported results may be overoptimistic and no other dataset has been used for validation. These works are most at risk of encountering generalization issues, and would benefit most from re-validation.

For the works listed under "Unknown: KAt Only", there are no other reported results, but it is not clear if the reported results would be susceptible. The seventeen works listed under "Mixed Parts: KAt and Other" and "Unknown: KAt and Other" have reported results on at least one other dataset. While the KAt results may be susceptible to data leakage, there are other results which can be referenced. Only the works listed under "Separate Parts" should be unaffected.

Random splits of KAt are often used for model validation, but a high accuracy for random split does not mean that the model has learned to identify fault conditions. Nor does a high accuracy on Bearing ID classification indicate the model has learned to identify the fault conditions of interest. In either of these cases, there is a risk of the model over-fitting to the specific parts, leading to generalization issues.

At this time, it is not known if the type of domain gap present in KAt is present in other datasets and it can not be determined in most cases, as there is often only one part available as an example of each class. Datasets collected with less sensitive sensors, or other types of sensors, may not include the additional information that serves to identify specific parts. However, without a wider variety of data, is it difficult to estimate the scope of the issue.

The accuracy drop that models often encounter when transferred from the Real Damages subset to the Artificial/Real split is often attributed to the differences between real damage and artificial damage [48]. However, the accuracy drop may not be due to the real-artificial difference but also, or in addition to, the individuality of the artificially damaged bearings.

D. SUMMARY

The recommendations are as follows:

- New data collection should include multiple parts in each condition of interest (including healthy, if applicable). Parts should be split between train and test sets.
- Train/test splits on the KAt dataset should ensure that the parts (Bearing IDs) are not shared between train and tests sets. Bearings IDs should not be used as class labels.
- Use reproducible train/test splitting methods on benchmark datasets.
- Use supervised techniques over unsupervised techniques in new FDD methods. Unsupervised techniques may tend to preserve sources of variance unrelated to the class.

Suggestions:

- Let test setups have a long rest period between collection periods (hours, days, etc.).
- Use more than one train/test splitting method.
- Filtering out high frequency information may improve generalization.
- When working an uncommon subset/split of a benchmark dataset to validate a technique, consider running a fast, simple classifier (kNN, RF, etc.) alongside to demonstrate the complexity of the problem.

VI. CONCLUSION

Despite the excellent performance of the examined methods on the CMHT Dataset with A_1, it can *not* be concluded that any of the diagnosis methods included in this paper have been able to successfully identify the fault conditions. The conclusion is that there exists domain shifts within the vibration data which enables data leakage and can produce vastly overoptimistic results. While results from mixed data may *look* good, it is important not to forget the problem at hand, and the restrictions that apply for practical applications. This paper highlights the issue of data leakage in this area, focusing on how the evaluation of model-free ML based FDD methods may be severely impacted by this phenomenon, which may impact other applications outside bearing fault diagnosis that use vibration data from rotating machines.

As the goal is to move towards more practical FDD methods, for use in industrial and consumer applications, systems and environments will only become more complex, which will likely increase the risk and severity of any domain shifts appearing. On the other side, as the objective of FDD is often to detect lower severity fault conditions, more sensitive sensors are required, which run the risk of picking up even more extraneous information. As the problem is to identify *fault conditions*, generalization is a requirement, therefore models should be *agnostic* to minor irrelevant differences within the data, no matter their cause. Thus, it is important to test for robustness and performance by *validating* models against new data sources such as using multiple parts for each class, swapping other parts, using multiple machines, etc.

Unfortunately, that is often a difficult solution due to cost and time constraints. It may be tempting to validate how models perform under *run-to-run* splitting, but it should not be assumed that those results will generalize. Furthermore, specific reporting of train/test splits is encouraged, and avoidance of random methods is preferable to enable reproducible results. Future work in this area may include expanded data collection to get a better understanding of the problem, testing existing models and methods for robustness against data leakage, techniques to detect and/or measure domain shifts within vibration data, an examination of domain shifts observed within a population of healthy machines, the impact of repairs and/or part swaps on domain shifts, and/or the severity of domain shifts in other types of sensors.

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